



UMD Wildlife Airport Strike StoryMap and Interactive Map

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Our Goal: To assist United States Department of Agriculture's (USDA) Animal and Plant Health Inspection Service (APHIS), Wildlife Services (WS), and Airport Wildlife Hazards Program (AWHP) to tell their story to the public; efforts mitigating wildlife hazards to aircraft for civilian and military airfields.



PROJECT GOAL

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Task 1

Task 1: Pinpoint locations of non-listed airports



Task 2

Task 2: Collect/analyze wildlife strike data



Task 3

Task 3: Compile strike data in an interactive map



Task 4

Task 4: Create dashboard enable interaction



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Task 1

Conduct Background Research on Wildlife Strikes and work being done to address them



Task 2

Design and Create StoryMap presenting research



Task 3

Incorporate GIS Dashboard into StoryMap



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Methods: GIS Map and Dashboard

1. Primarily accomplished through ESRI's ArcGIS online (AGOL) and Dashboards applications
1. Created map displaying all airports associated with Airport Wildlife Hazards Program
1. Filtered by airport name, use



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Methods: Dashboard

Filtering Data

From *Table 2* of the written report

Index Number	Repair Costs
Incident Date	Other Costs
Incident Month	Inflation Adjusted Repair Costs
Incident Year	Inflation Adjusted Other Costs
Time of Day	Indicated Damage
Airport ID	Damage Level
Airport Longitude	Effect
Airport Latitude	Species
Airport Name	Remarks
State	Warned
FAA Region	Number Seen
Aircraft Type	Number Struck
Sky	Size
Precipitation	Number of Injuries
AOS	Number of Fatalities

PROJECT GOAL

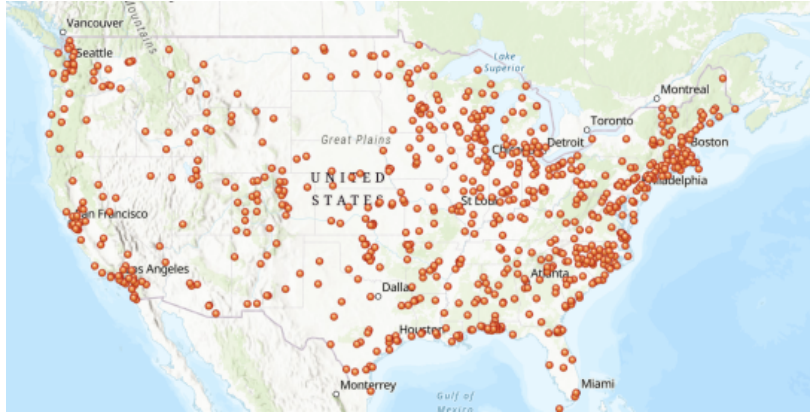
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FINDINGS

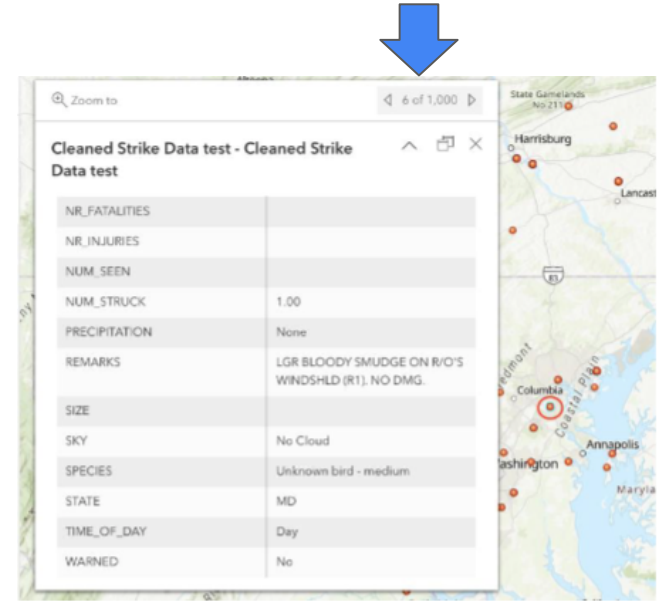
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RECOMMENDATIONS

Methods: Interactive Map



The interactive map zoomed in to highlight the US



From Figure 1 in the written report

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total damages

Total In Damages

861.85M

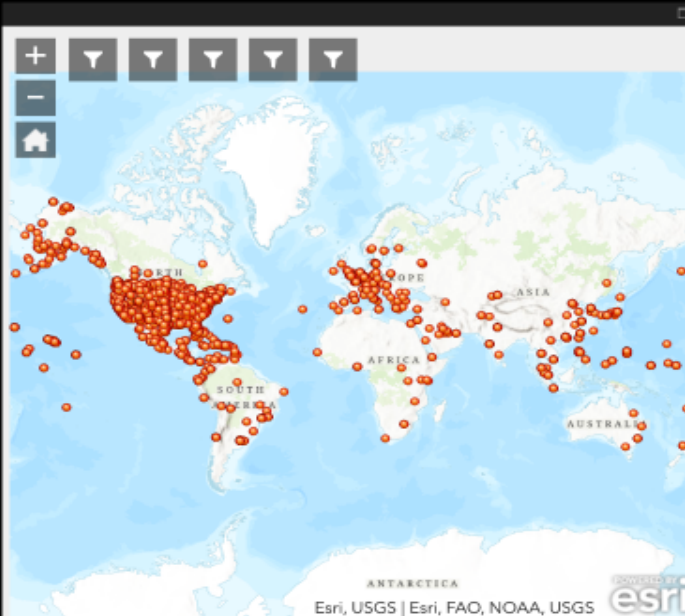
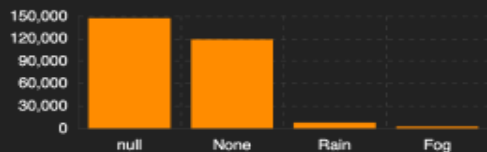
time of strike

Time of Strike



Precipitation

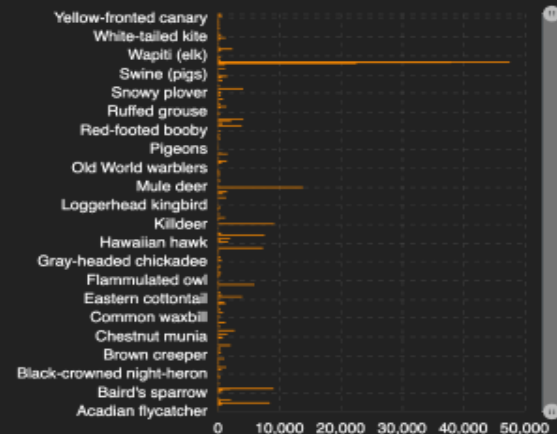
Precipitation



Esri, USGS | Esri, FAO, NOAA, USGS

Species

Species Involved



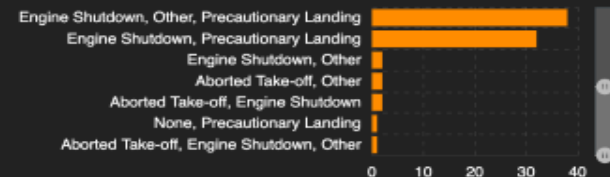
injuries

Injuries

345

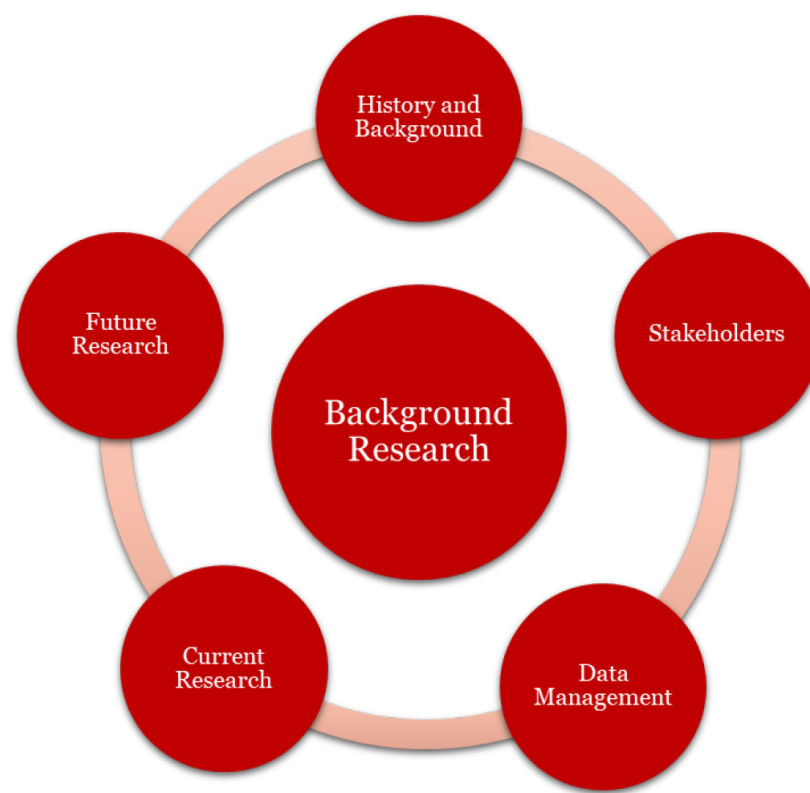
Effect

Effect of Strike

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Methods StoryMap



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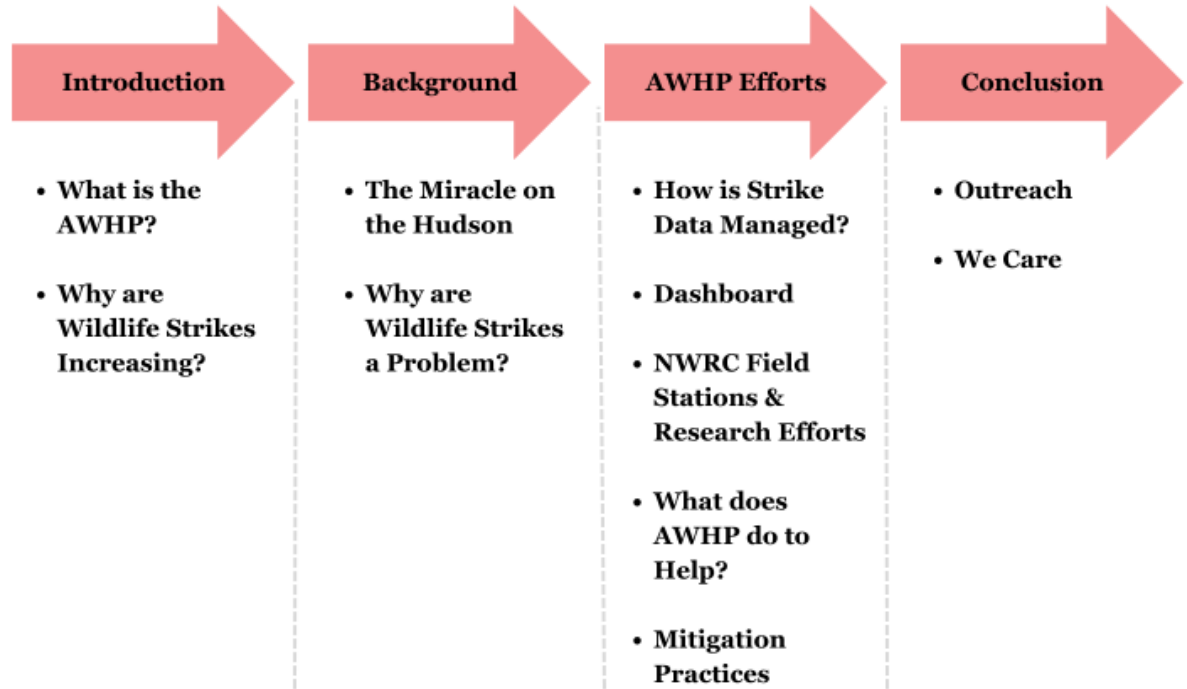
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StoryMap Methods



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total damages

Total In Damages

861.85M

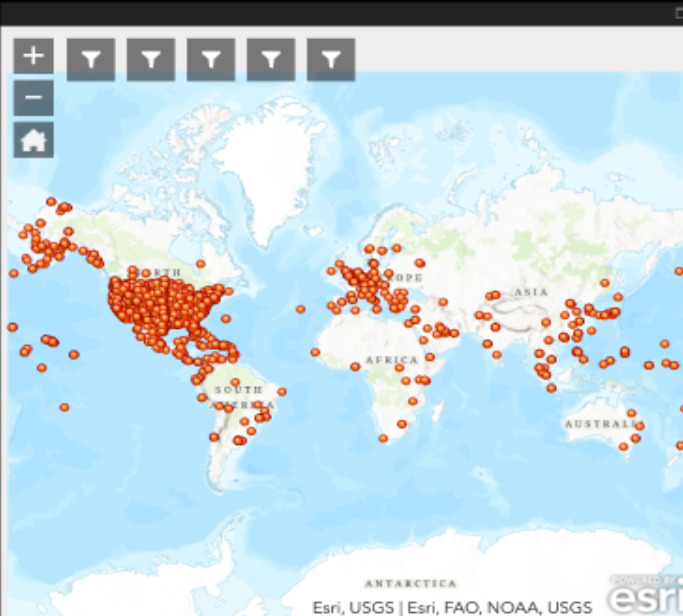
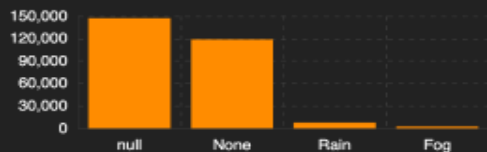
time of strike

Time of Strike



Precipitation

Precipitation



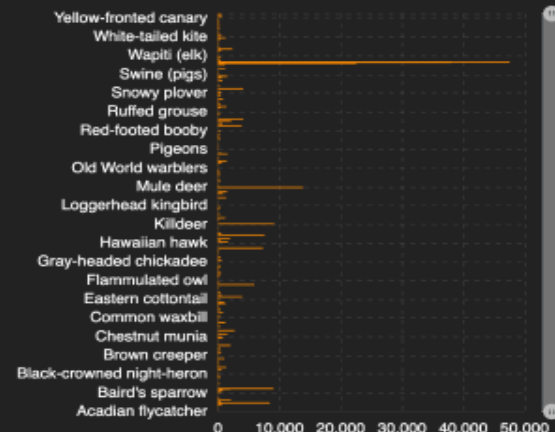
injuries

Injuries

345

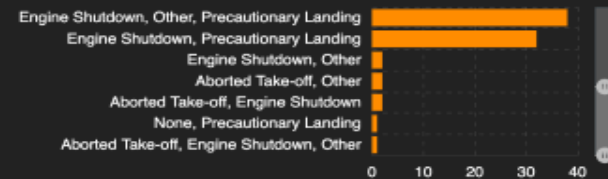
Species

Species Involved



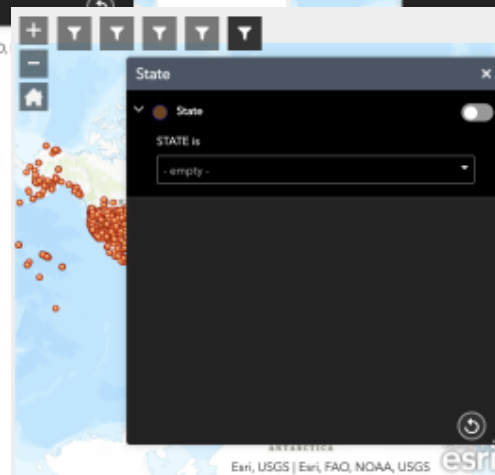
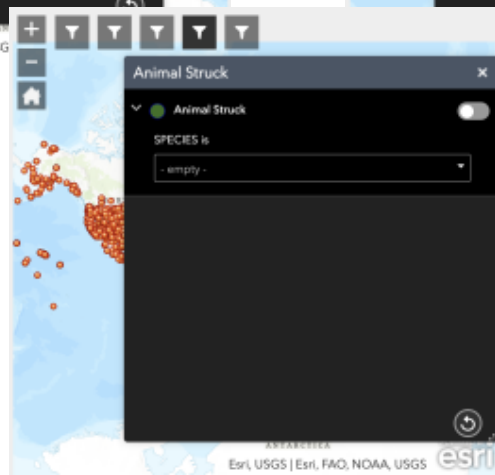
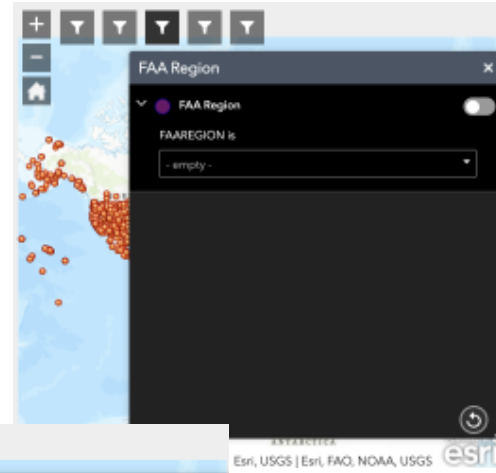
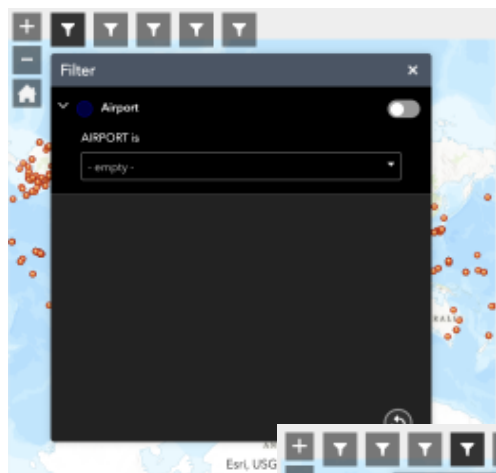
Effect

Effect of Strike



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Dulles International Airport (IAD)

1990-2023, Data Credit FAA Strike Database: <https://wildlife.faa.gov/home>



Costs of Repairs

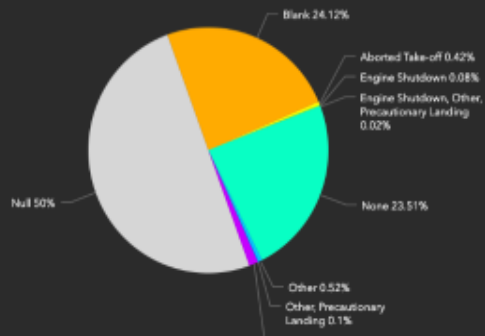
17.2M

Costs of Strikes (Other)

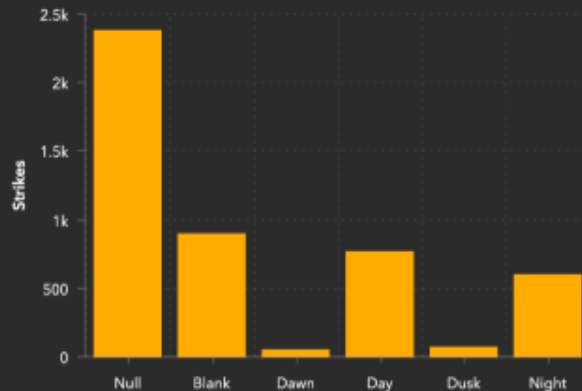
313.2k

Strikes

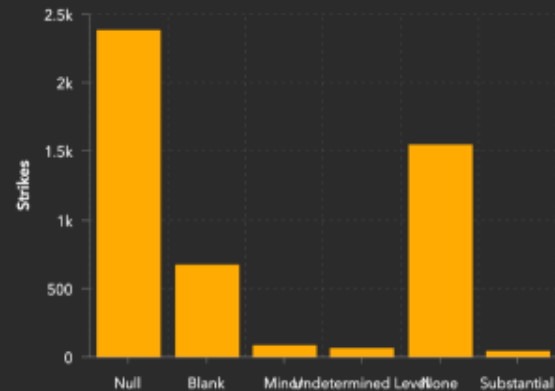
4.8k



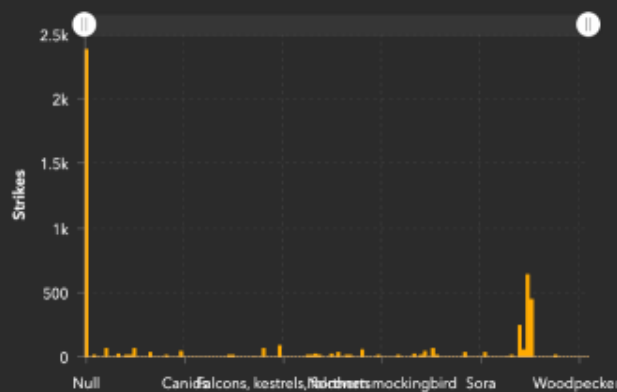
Time of Day



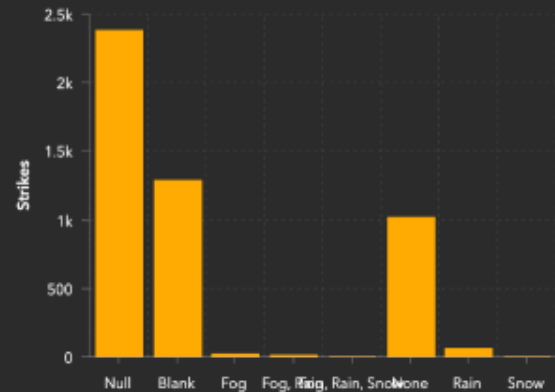
Damage Level



Species

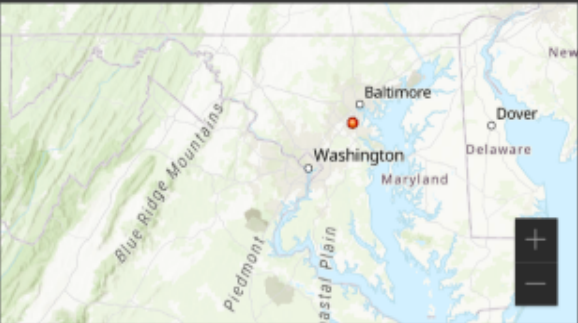


Weather



Baltimore/Washington International Thurgood Marshall Airport (BWI)

1990-2023, Data Credit FAA Strike Database: <https://wildlife.faa.gov/home>



Costs of Repairs

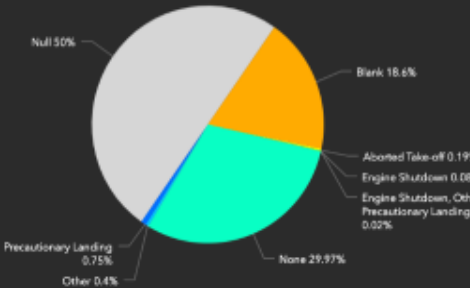
3.1M

Costs from Strikes (Other)

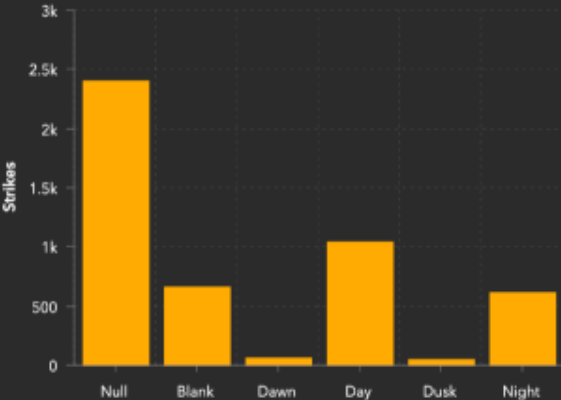
842.2k

Strikes

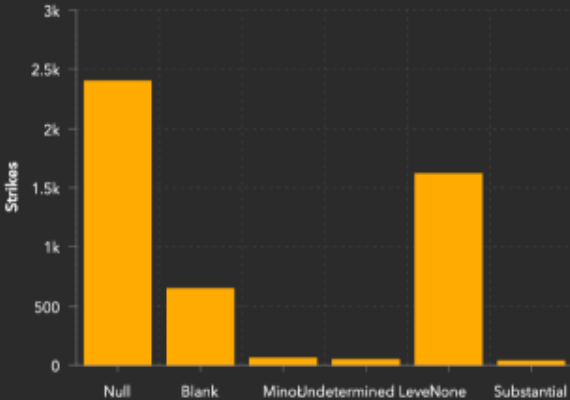
4.8k



Time of Day



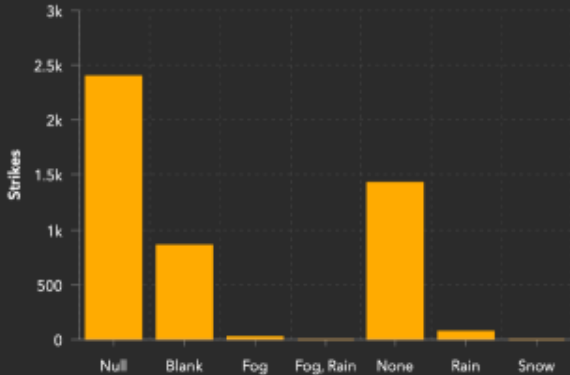
Damage Level



Species



Weather



Ronald Reagan Washington National Airport (DCA)

1990-2023, Data Credit FAA Strike Database: <https://wildlife.faa.gov/home>



Esri, USGS | Esri, FAO, NOAA, USGS Powered by Esri

Costs of Repairs

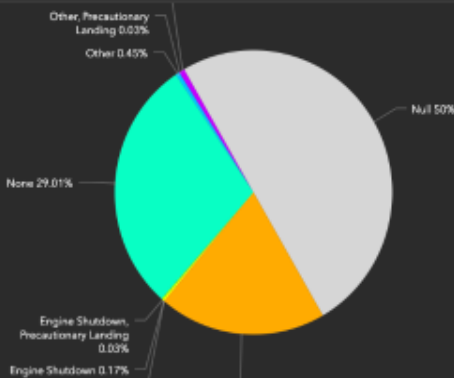
7.1M

Costs from Strikes (Other)

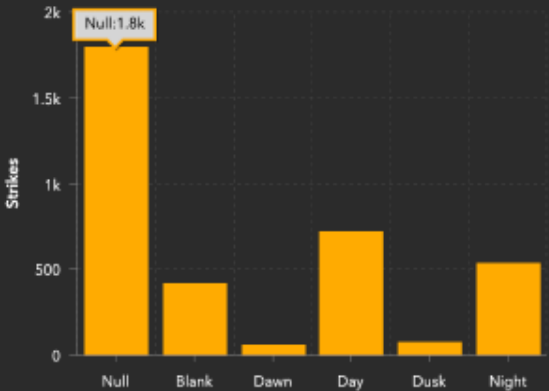
1.7M

Strikes

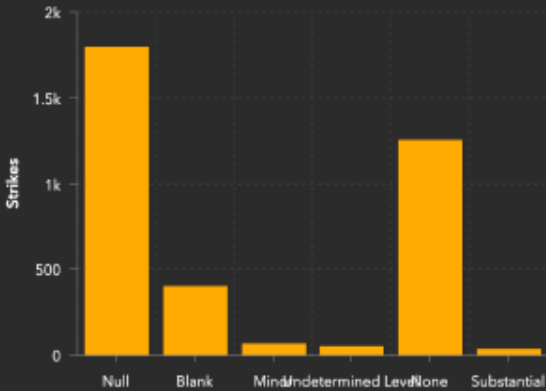
3.6k



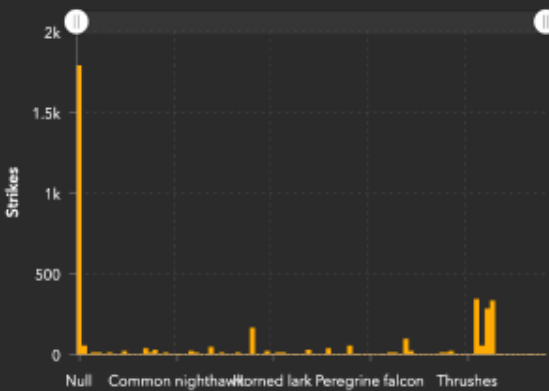
Time of Day



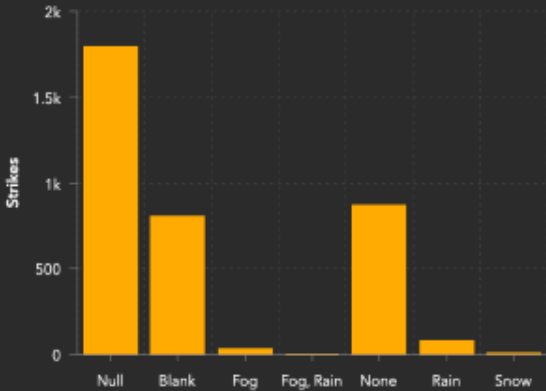
Damage Level



Species



Weather



Limitations



ArcGIS StoryMaps

Storytelling that resonates

Transform your digital storytelling with custom maps.



ArcGIS Online

Mapping and analysis: location intelligence for everyone

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Recommendations



“ It's a personal legacy as a grandkid of a cold war pilot when wildlife strikes weren't often considered. Today, we get to focus on our work to change the narrative for Aviation in a meaningful way.



“ I care about the safety of the men and women serving our country, particularly those carrying out the flying missions; as well as those involved with air operations.



“ Working towards positive outcomes for both people and wildlife is a big motivating factor for my work to reduce aircraft/wildlife strikes.

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Questions?



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